

In-hand cylinder reorientation with a delayed target

xArm7 + LEAP hand in MuJoCo 3.13. Scripted grasp, learned in-hand reorientation, target revealed at lift.

Result. The full task runs end to end: grasp, lift, target revealed, in-hand reorientation, 2 s hold. It succeeds in **8.5 %** of 200 episodes (17/200, 95 % Wilson CI 5.4–13.2 %), against **0.5 %** (1/200) for the same grasp followed by holding still, on identical objects and targets. The grasp is the ceiling: 40 % of episodes never reach reorientation. Of those that do, the policy succeeds in 13 % (0 % for holding still). This is below the 50 % bar; we report where it breaks rather than narrowing further.

Scope actually measured. Lying rods, diameter 4.4–5.6 cm, height 8–12 cm (α 1.4–2.7), density 500–900 kg/m³. Targets: axis within 40° of palm y, position in a 3 × 3 × 0.8 cm box on the palm. Outside that we report measured rates, not claims (§6).

System (same 200 episodes, seeds 123–126)	handover	success	success handover	median e_{pos}	median e_{ang}
Scripted grasp + hold still (baseline)	60.0 %	0.5 %	0 / 120	3.6 cm	35°
Scripted grasp + learned reorient	60.0 %	8.5 %	16 / 120	3.3 cm	25°

Errors are at episode end over handed-over episodes. Success needs $e_{\text{pos}} \leq 1.5$ cm and $e_{\text{ang}} \leq 15^\circ$ held for 2 s with the arm frozen and hand-only contact (thresholds as specified, not loosened).

1. What we chose not to do, and why

- **One shape band, one start pose.** A same-morning, roughly three-hour budget. Getting the full pipeline honest on one band beats a wide envelope we cannot measure. The wider envelope is swept in eval only (§6).
- **No learned grasp.** The grasp is a script. A learned grasp is the obvious next step. Its reward cannot see T^* (it is hidden until lift), so the right objective is the reorient critic's value averaged over the target prior. That is implemented (plan A in the repo) but not trained here.
- **No perception stack.** Ground-truth pose stands in for vision, gated by visibility (§2).
- **Plans A, B, C in the repo are not the submission.** They share this core but were not trained to convergence. Only the scripted-grasp + learned-reorient system (PlanS) is evaluated.

2. Assumptions

Item	Choice
Robot	UFactory xArm7 + LEAP right hand (MuJoCo Menagerie), hand on a 40 mm bracket past the flange. 23 position actuators with Menagerie gains and ctrl ranges; collisions on; joint limits respected.
Palm frame	Origin on the palm face under the finger bases; x toward the fingertips, y across the fingers, z the outward palm normal.
Object	Uniform-density cylinder. r 2.2–2.8 cm, h 8–12 cm i.i.d., density 500–900 kg/m ³ (mass 60–270 g). Finger span $\approx$ 9 cm, fingertip reach $\approx$ 12 cm, so the rod is about half the span wide and about one span long.
Start pose	Lying on its side, uniform yaw, x 0.36–0.56 m, y $\pm$ 0.14 m in front of the robot base.
Friction	Object μ 0.7–1.1, ground μ 0.5–0.9, sampled per episode (sliding; MuJoCo default torsional/rolling).
Contact model / solver	2 ms step, implicit-fast, elliptic cones, impratio 10, 2 noslip iterations. Object geom condim 4, solref (10 ms, 1), solimp (0.9, 0.95, 2 mm). The 4 ms default was numerically stable but

	not accurate (§5).
Control	25 Hz, delta joint-position targets, no latency. Grasp: arm and hand. Reorient: the arm holds the palm-up pose and fingers move at most 0.1 rad per step.
Tactile	Five pads (four fingertips, palm), read at 25 Hz: contact flag, summed normal force, force-weighted contact centroid in the pad frame. Noiseless. Counts every contact, since a real pad does not know what it touches. Comparable to a Xela uSkin patch per link.
Cameras / vision gate	Two fixed cameras at (1.1, ± 0.7 , 0.9) m and one wrist camera on a bracket behind the palm. The stand-in exposes ground-truth object pose + (r, h) at 5 Hz only when $\geq 30\%$ of 32 area-weighted surface samples are unoccluded from at least one camera (ray-cast). Otherwise the last estimate is held with the flag at 0. 30 % is about the silhouette a pose estimator needs for a textureless cylinder of known class.
Deployment observation	Arm/hand q and dq, previous action, current targets, tactile, palm pose from kinematics, gated vision, target block (zeros and flag 0 until lift), time. No object pose while occluded, no mass/friction, no object-ground forces.
Lift detector	At each control step (25 Hz), contacts are classified from MuJoCo contact geoms: lift is the first step where the object touches ≥ 1 hand geom and no floor, arm-link or other geom. The world then writes T^* into the observation.
T^* sampling and feasibility	Drawn once per episode, fixed in the palm frame, independent of the grasp. Axis: palm y rotated by an angle uniform in $[0^\circ, 40^\circ]$ about a random perpendicular. Position: $x \in [-4.5, -1.5]$ cm, $y \in [-1.5, 1.5]$ cm, z = palm-contact height for that axis + $[0.2, 1.0]$ cm. Feasible = within 8 cm of the palm origin and no penetration with the palm/wrist at open fingers (resampled otherwise). The box was centred once on the median handover pose over 200 grasps; the hold-still baseline measures how much of it is free.
Termination	Success; object contact with anything but the hand after lift (violation); 5 steps without hand contact after lift (drop); object > 0.5 m from the workspace centre; 20 s horizon.
Stage transition	Grasp $\rightarrow$ reorient when the lift flag is set <i>and</i> the arm has reached the palm-up hold configuration (joint error < 0.03 rad), or 200 steps after lift.

3. Decomposition and stage interfaces

Stage 1: scripted grasp (`inhand/planc/scripted_grasp.py`). It reads only the deploy observation. The object pose comes from the gated vision stand-in, which sees the rod on the floor.

1. IK to a palm-down pre-grasp above the rod, fingers across its axis, rod 5 cm toward the fingertips from the palm origin.
2. Descend until the palm is 2 cm *into* the rod top, which pins it to the floor.
3. Close the fingers over 12 steps.
4. **Catch:** if the lift flag flips during the close (the rod popped), finish the close and rise at 30 cm/s.
5. Lift 15 cm at 10 cm/s, then roll the wrist to palm-up in joint space.

It never reads T^* . It uses the lift flag only as an event.

Stage 2: learned reorient (`PlanS` in `inhand/systems.py`). A recurrent PPO actor on the deploy observation, running from handover to the end of the episode. The interface between the stages is the simulator state itself: the policy is trained from a bank of 266 states saved at real handovers (`scripts/collect_handovers.py`), mixed 70/30 with synthetic in-hand drops for coverage.

4. MDP of the learned component

Observation	Actor: the 161-d deploy vector above, with time measured from handover. Critic: deploy vector + object pose/velocity in the palm frame, r, h, mass, frictions, contact flags, true target (asymmetric
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actor-critic, train only).

Action	16 finger delta-targets in $[-1, 1] \times 0.1$ rad. Arm actions are zeroed: the target is palm-frame, and moving the arm only adds ways to touch the object with an arm link.
Reward	$\exp(-e_{\text{pos}}/3 \text{ cm}) + \exp(-e_{\text{ang}}/0.4 \text{ rad}) + 2 \cdot [\text{in tolerance}] + 50 \cdot [\text{success}] - 10 \cdot [\text{drop or violation}] - 0.002 \ \text{all} \ ^2$. All shaping is positive and bounded. With negative per-step error the first policies learned that ending the episode early (throwing the rod into the arm) was optimal.
Termination	As §2, 10 s horizon (the grasp uses about 9 s of the 20 s episode).
Algorithm	PPO, GRU(256) over MLP(512, 256, 128). γ 0.99, λ 0.95, clip 0.2, 4 epochs $\times$ 4 minibatches, KL-adaptive LR (target 0.02), 56 envs $\times$ 64 steps per iteration.
Why	The fingers occlude the rod, so pose must be carried through time from tactile and proprioception: hence a recurrent actor. Privileged state makes the value function learnable in minutes rather than hours. Training directly on deploy observations removes the teacher–student distillation step, which the time budget could not afford.

5. The grasp stage: what it costs and what fixed it

The grasp decides both whether the episode gets to reorientation and where the object starts in the hand. Rates below are handovers (object hand-only after the palm-up roll) on lying rods of the trained band.

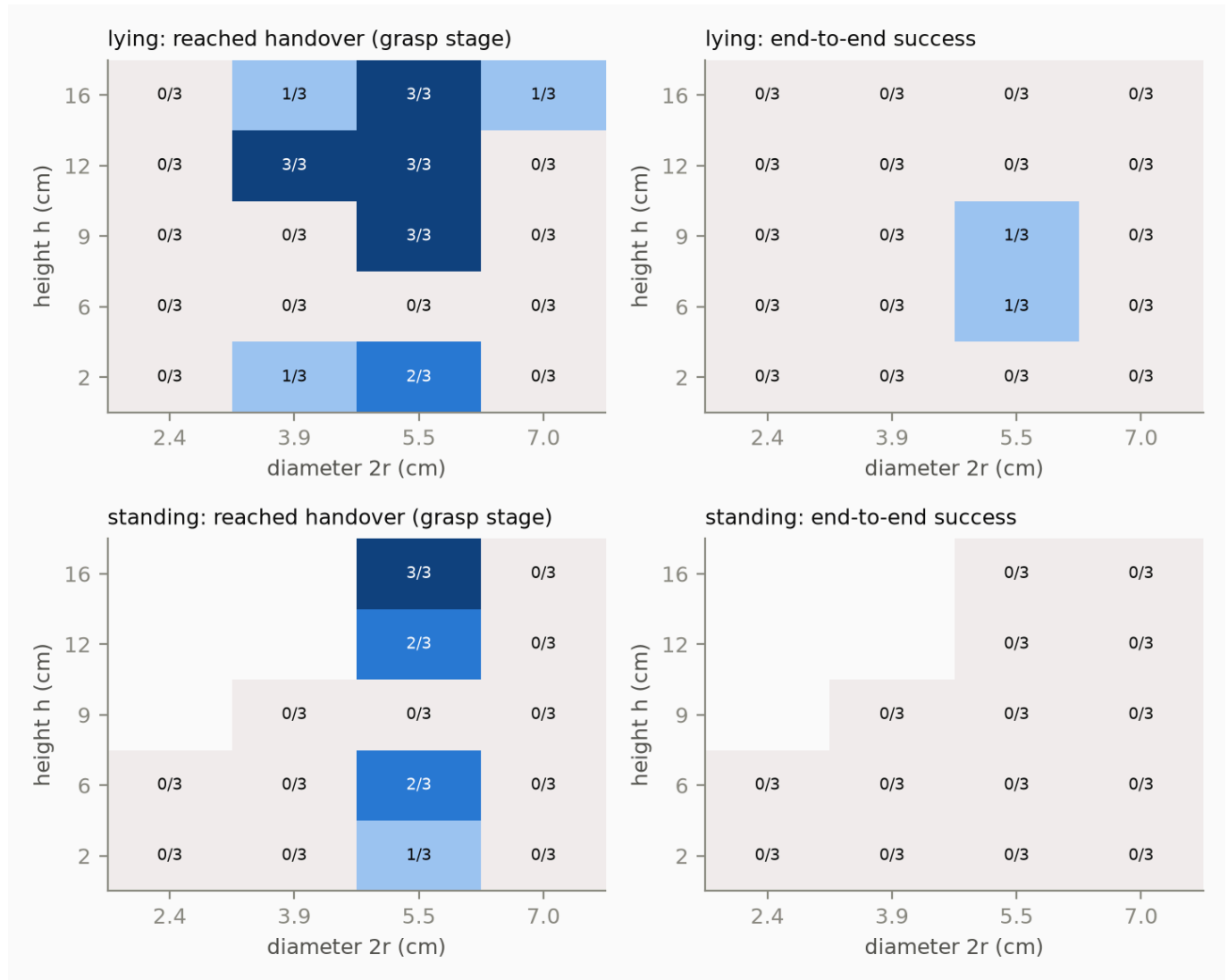
Change	handover	Evidence
Original script, 4 ms step, impratio 100, stiff object contact	1/12	Every failure: a finger touches the rod, the rod leaves at 0.5–0.8 m/s in one control step, which counts as lift, then lands (violation).
Sweeps at 4 ms: offset, clearance, close order, compliant close (stall limit), palm pressed into the rod	0–25 %	No setting fixed the flick; stall-limited fingers never wrapped. Best: palm pressed 2 cm (4/16).
Original script at 1 ms / palm-pressed at 1 ms	5/16, 8/16	The flick was largely a time-step artefact.
Palm-pressed, 2 ms, solref 10 ms, solimp width 2 mm, impratio 10	8/16	Same as 1 ms at half the cost; adopted. 12/24 on a second seed.
+ fingers close over 12 steps instead of 25	34–53 %	32 episodes each on two seeds. Remaining failure: the close still pops the rod.
+ catch on lift flag	53–66 %	32 episodes each on two seeds.
Final script, 480 episodes	55 %	266 handovers saved as the reorient reset bank.

Trade-off. A scripted grasp took about 40 minutes to reach 55 % and needs no training, but it caps end-to-end success at 55 %. It also hands over a spread of in-hand poses (axis 2° – 59° from palm y, 10th–90th percentile) that the reorient policy must absorb. A learned, value-aware grasp could both lift more reliably and pick grasps that are easy to reorient from. That is the single change with the largest expected gain.

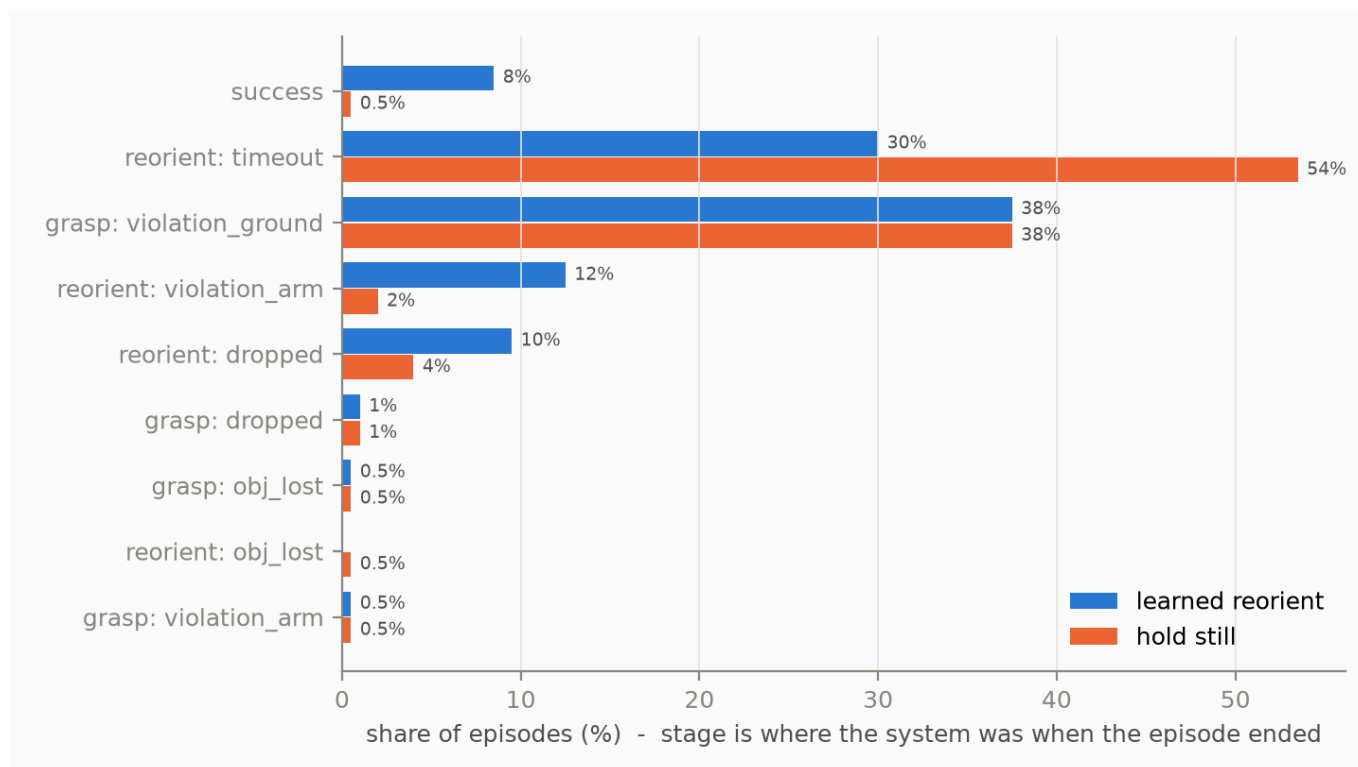
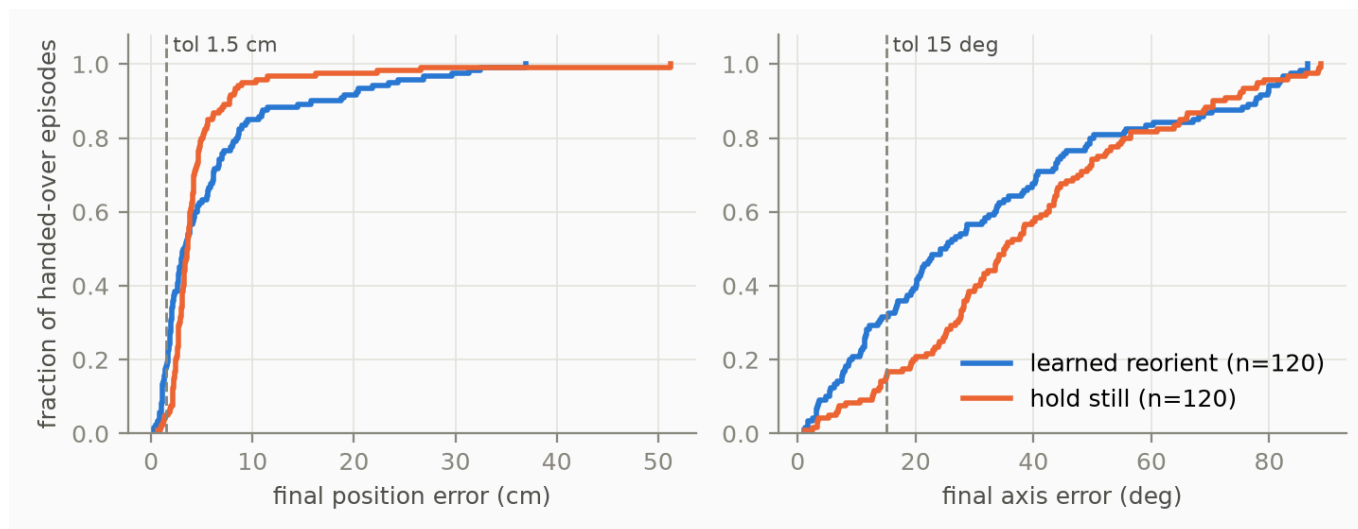
6. Envelope

The policy was trained on the narrow band only. We swept the WIDE envelope in eval: diameter 2.4–7.0 cm and height 2–16 cm on a 4×5 grid, 3 objects per cell, both start poses (standing only where $\alpha \leq 3$; tall standing rods are tipped over first). 105 episodes.

- **Grasp envelope.** Handover in 24 % overall. Lying: 17/60, concentrated at 5.5 cm diameter ($h \geq 9$ cm: 9/9) and 3.9 cm at $h = 12$ cm. Standing: 8/45, only at 5.5 cm. Thin rods (2.4 cm) never hand over (0/21): the hand pushes them out of the workspace or never gets them off the floor. Fat rods (7.0 cm) do not fit the wrap and pop off the floor (22/30 ground violations).
- **End-to-end.** 2/105 (1.9 %). Both successes are lying 5.5 cm rods ($h = 6$ and 9 cm), inside or next to the trained band; the $h = 6$ cm one matched T^* during the roll, before handover. Nothing outside the trained diameter succeeded. Within the band, longer rods did better: $\alpha = 2-3$ succeeded 11/103 vs $\alpha = 1-2$ 6/97.
- **Measured range.** Lying rods, diameter 4.4–5.6 cm, height 8–12 cm (the trained band), success 8.5 %; the grid adds successes at 5.5 cm $\times$ 6 and 9 cm. We do not claim standing starts or other diameters.



7. Errors and failure taxonomy



Rates over the 200 paired episodes, attributed to the stage the system was in when the episode ended.

Failure	Stage	learned	hold	Why
Object hits the floor after lift	grasp	37.5 %	37.5 %	The close pops the rod off the floor (this counts as lift) and the catch misses it. Identical in both columns because the grasp is shared.
Timeout, in hand but out of tolerance	reorient	30.0 %	53.5 %	The policy moves the rod toward T* but stalls. Median final error of handed-over episodes 3.3 cm / 25° (hold 3.6 cm / 35°). Axis tolerance is met in 32 % and position in 18 % of handovers: position is the harder half.
Object touches an arm link	reorient	12.5 %	2.0 %	Caused by the policy. The target box sits toward the wrist (where grasps leave the rod), and rolling the rod that way reaches the wrist bracket.
Dropped (5 steps without hand contact)	reorient	9.5 %	4.0 %	Caused by the policy: moving the rod costs grip, and the policy has not learned to keep a stable contact set while it moves.

Dropped / lost / arm during grasp	grasp	2.0 %	2.0 %	Slip during the wrist roll.
Success		8.5 %	0.5 %	One success per column happened before handover: the rod already matched T* during the roll.

Per-stage attribution: grasp stage 40 % of episodes lost, reorient stage 52 %, success 8.5 %. Two fixes would each roughly double the result: removing the grasp pop, or eliminating the policy-induced arm contacts and drops.

8. Limitations and next steps

- **Grasp is the ceiling.** 40 % of episodes never reach reorientation. Next: train the value-aware grasp (plan A) on the 2 ms physics, warm-started from the script by DAgger.
- **Narrow band.** One shape band and one start pose. Next: widen r and h with ADR once the grasp is learned. Standing starts need either a tip-over or a side grasp.
- **Target prior centred on the grasp's typical pose.** A defensible feasibility choice, but the policy only has to tilt $\leq 40^\circ$ and move a few cm. Next: widen the cone with the existing curriculum.
- **Compute.** Single laptop, CPU MuJoCo. MJX or MuJoCo Warp would give roughly 100× the environment steps.
- **Vision stand-in.** Ground truth when ≥ 30 % visible. A real estimator (e.g. FoundationPose) would add latency and error that we have not modelled.

9. Compute

All training and evaluation on one laptop CPU (Apple M5 Max, 18 cores), CPU MuJoCo, no GPU.

- **Reorient policy:** 80 min, 56 envs over 14 processes, 7.55 M env steps at 1,570 steps/s. The submitted checkpoint is the best-smoothed-success snapshot at 5.53 M steps (57 min). It was chosen over the final checkpoint on 120 held-out validation episodes (seeds 900–903): 10.8 % vs 7.5 % end-to-end. The evaluation seeds (123–126) were not used for selection.
- **Abandoned runs:** about 12 min / 1 M steps, before the physics fix, the handover bank and re-centring the target box.
- **Grasp sweeps:** about 25 min on 8 processes, 16–32 episodes per setting.
- **Evaluation:** about 5 min on 13 processes.
- **Cluster:** NCSA Delta jobs were queued as a hedge but never started before the deadline.

Wall clock from start to report: about 2.7 h.

10. References and tools

MuJoCo 3.13 and MuJoCo Menagerie (xArm7, LEAP hand); PyTorch; custom PPO following rl_games' KL-adaptive schedule. Pinto et al., Asymmetric Actor Critic (2017); Chen et al., Visual Dexterity (2023); Handa et al., DeXtreme (2023); Andrychowicz et al., HER (2017); Ross et al., DAgger (2011).